

Total

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1. Analyze the differentiability of $y = \sqrt[5]{x^4}$. Show work algebraically and then explain.

$$\frac{dy}{dx} = \frac{d}{dx} \left(x^{\frac{4}{5}} \right) = \frac{4}{5} x^{-\frac{1}{5}} = \frac{4}{55x}$$

rationalized version = $\frac{4\sqrt[5]{x^4}}{5x}$

As you can see, if $x = 0$, $\frac{dy}{dx} = \frac{4}{0}$ which is undefined, and because this is an odd root, negative numbers for x exist. $\therefore$ Function is differentiable for all values of x , except $x = 0$

Also, this is a cusp at 0

2. Comment on the validity of the following statement: "If $p'(x) = q'(x)$, then $p(x) = q(x)$ "

This is false, this is because the derivative practically ignores all constants (they become 0) Example: $p(x) = x^2 + 2$ $q(x) = x^2$, $p'(x) = 2x$ $q'(x) = 2x$
In this example, the derivatives are equal but the original functions are not because of the difference in constants

3. Differentiate each of the following. Be sure to use the appropriate notation. Fully simplify so that no rational nor negative exponents remain in final answers.

a) $f(x) = -4x^2 + \frac{2}{x^2} - \pi^2$

$$f'(x) = -8x + 2x^{-2} - \pi^2$$

$$f'(x) = -8x - 4x^{-3}$$

$$= \frac{-8x - 4}{x^3}$$

b) $y = 3\sqrt{x}(2 - 5\sqrt{x})$

$$y = 3x^{\frac{1}{2}}(2 - 5x^{\frac{1}{2}})$$

$$= 6x^{\frac{1}{2}} - 15x$$

$$\frac{dy}{dx} = 3x^{-\frac{1}{2}} - 15$$

$$= \frac{3}{\sqrt{x}} - 15$$

$$\pm \frac{3\sqrt{x}}{x} - 15$$

c) $f(x) = \frac{10x^2 + 1}{5x}$

$$f'(x) = \frac{20x}{5x} + \frac{1}{5x}$$

$$= 2x + (5x)^{-1}$$

$$f'(x) = 2 + (5x)^{-2}$$

$$= 2 - \frac{1(5)}{(5x)^2}$$

$$= \frac{2 - \frac{5}{25x^2}}{25x^2} = \frac{2 - \frac{1}{5x^2}}{25x^2}$$

5. Determine the point(s), if any, where the tangents to the curve $y = 6x^3 + 2x^2$ are parallel to the line $5x + y - 6 = 0$. Justify your response.

$$-y = 5x - 6$$

$$y = -5x + 6$$

$$\downarrow$$

$$-5 = 18x^2 + 4x$$

$$0 = 18x^2 + 4x + 5$$

$$\frac{dy}{dx} = 18x^2 + 4x$$

$$x = \frac{-4 \pm \sqrt{4^2 - 4(18)(5)}}{36} = \frac{-4 \pm \sqrt{16 - 360}}{36} = \frac{-4 \pm \sqrt{-344}}{36} = \text{DNE}$$

imaginary



∴ There are no points because $18x^2 + 4x$ can't ever equal -5 the slope of -5 if using real numbers

6. Determine the value of c such that the tangent line to $f(x) = (2x^3 - 1)(3x^2 + c)$ at $x = 1$ is horizontal.

$$f(x) = 6x^5 + 2cx^3 - 3x^2 - c$$

$$f'(x) = 30x^4 + 6cx^2 - 6x - c$$

$$c = dx$$

$$\frac{d}{dx}(dx) = d$$

$$\frac{d}{dx}(c) = d$$

$$d = \frac{c}{x} = \frac{-24x}{5x} = \frac{-24}{5}$$

$$f'(1) = 0$$

$$0 = 30 + 6c - 6 - d$$

$$0 = 24 + 6c - d$$

$$0 = 24 + 6(d(1)) - d$$

$$0 = 24 + 6d - d$$

$$-24 = 5d$$

$$5d = \frac{-24}{5}$$

$$c = \frac{-24x}{5}$$

$$f'(1) = 30 - \frac{144}{5} - 6 + \frac{24}{5} = 0$$

8. Use first principles to determine the derivative function for $f(x) = \frac{3}{2x-1}$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{3}{2x+2h-1} - \frac{3}{2x-1}}{h} = \lim_{h \rightarrow 0} \frac{\frac{3(2x-1) - 3(2x+2h-1)}{h(2x-1)(2x+2h-1)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6x-3-6x-6h+3}{h(2x-1)(2x+2h-1)} = \lim_{h \rightarrow 0} \frac{-6}{(2x-1)^2} = \frac{-6}{(2x-1)^2}$$